

KarigarSetu

Voice-First Market Linkage for Marginalized Artisans

- ◆ **Problem Statement ID** — SIH26090
- ◆ **Problem Statement Title** — AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans
- ◆ **Theme** — Heritage and Culture
- ◆ **PS Category** — Software / Hardware
- ◆ **Team Name** — Renaissance Guild



◆ Empowering rural weavers and makers by converting regional spoken dialects directly into professional, global market-ready catalogues.

❖ Proposed Solution – Voice-First Multilingual Ecosystem & Feature Matrix

> CORE SOLUTION PILLARS & INNOVATIVE FEATURES <

Voice-First Multilingual App

Translates Telugu speech + raw photo into buyer-ready listings.



01

02



Zero-Friction Onboarding

Eliminates English forms; empowers rural non-tech artisans.

AI Core + Artisan-Approved HITL

Whisper ASR + SAM Vision + Llama-3 with mandatory human confirmation.



03

04



Heritage Craft Geo-Map

Interactive map locating rural art clusters & indigenous makers.

AI Space Decor & Visualiser

Photo-based room styling linking authentic craft items.



05

06



Multilingual Govt Scheme Finder

Matches artisan profiles to financial welfare subsidies.

Direct Artisan Chat & Orders

Direct channel for custom design negotiations & bulk orders.



07

08



Price History & AI Chatbot

Transparent price analytics & interactive voice guidance bot.

SOLUTION ARCHITECTURE: END-TO-END PIPELINE DEMONSTRATION

1. ARTISAN INPUT



📱 Unedited mobile photo

🔊 TELUGU VOICE NOTE (RAW)

“ఇది మూ చేతి పన్నవను మరల్లు, మేమ్మనూ ప బు రత్తద తలతలలే తయారు చేస్తాము. మా ఉల్లిదల్లు ఫోటోలు ఇవి. దయచేసి మంచి ధఫోడుం చేయండి.”

2. AI ENGINE CORE

- 1 Photo Enhance**
Auto-corrects poor lighting, isolates objects, & removes clutter via SAM vision.
- 2 Speech-to-Text**
Whisper ASR engine transcribes regional Telugu speech note to transcript.
- 3 Translation & Tags**
Translates Telugu to English; structures attributes into e-commerce tags.
- 4 Smart Pricing**
Recommends market price based on raw materials, labor hours & trends.

3. MARKET CATALOG



🌟 AI-Enhanced studio photo

Handwoven Bamboo Basket

Handcrafted in Medipally, TS

- Material: Natural Organic Bamboo
- Fair Price: ₹400 (Artisan Price)

✓ Artisan-Approved AI Draft



Artisan Input (Voice+Photo)



AI Processing Core



E-Commerce Catalog Listing



Global Market Linkage

KARIGAR SETHU - TECHNICAL APPROACH

Team Renaissance Guild

METHODOLOGY & WORKFLOW

1. Input Ingestion -
Camera & Audio



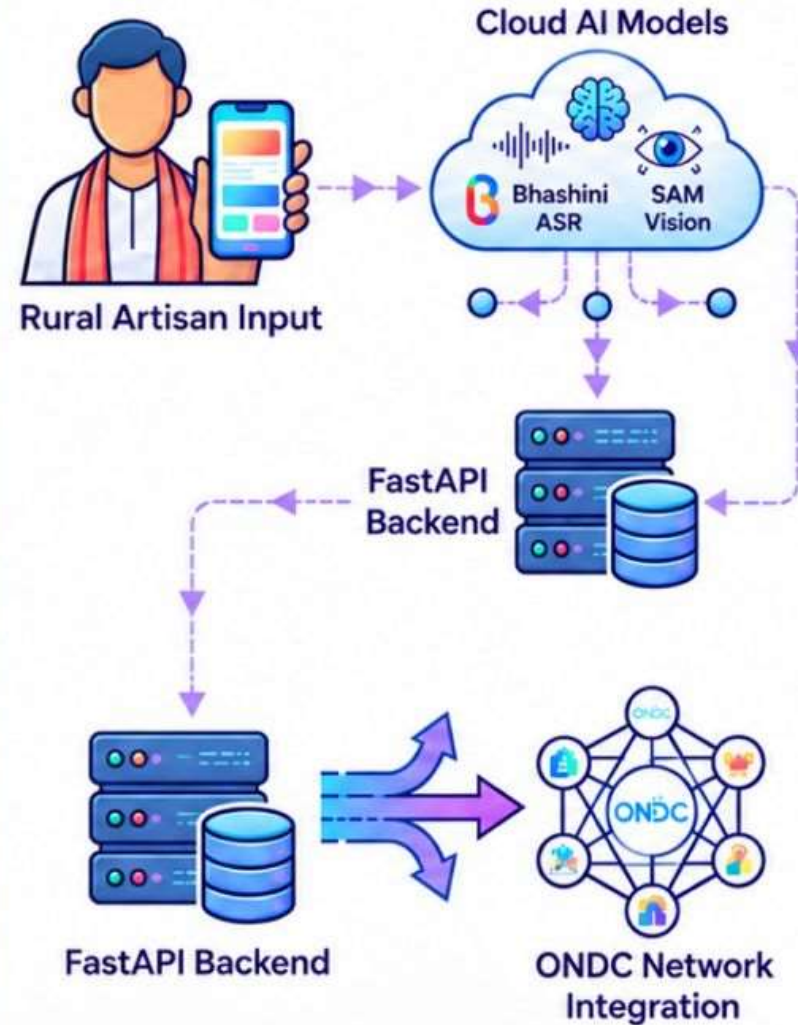
2. Dual AI Pipeline -
SAM Vision & Whisper ASR



4. HITL Approval -
ONDC Schema



3. Translation & LLM
Structuring - Llama-3



PROPOSED TECH STACK



FEASIBILITY, VIABILITY & BUSINESS POTENTIAL

FEASIBILITY ANALYSIS



Edge-AI Background Removal



Bhashini Multilingual Voice Input
Telugu/Hindi



Offline Syncing & Low Digital Literacy UI



VIABILITY & RISKS



Human-in-the-Loop Review



Fair AI Price Guidance



ONDC Network Integration



Phased Craft Cluster Rollout

BUSINESS POTENTIAL



Cuts Middlemen Commissions



Direct B2B Wholesale Market



Rural Female Empowerment & Fair Revenue



❖ Impact & Benefits

FOR MARGINALIZED ARTISANS

- Professional Digital Cataloguing: Creates a stunning, standardized digital product listing directly from a raw smartphone photo and regional voice inputs.
- **Complete Language Inclusion: Reduces critical dependency on English writing, advanced computer literacy, and expensive professional agency services.**
- Less Onboarding Effort & Friction: Enables wider buyer discovery. Eliminates physical-only sales limitations at local fairs and seasonal exhibitions.
- **Total Control & Full Agency: Keeps artisans firmly in control. AI-generated product descriptions stay fully editable and require confirmation before publishing.**

SEAMLESS PRODUCT JOURNEY

1. Artisan Input

Raw Photo + Telugu Voice Note

▼

2. AI Engine Draft

ASR Transcription & Listing Generation

▼

3. Artisan Verification

Review, Edit, and Tap to Approve

▼

4. Buyer-Ready Catalog

Live Listing matching with Bulk Buyers

FOR BUYERS & ECOSYSTEMS

- Seamless Buyer Discovery: Provides large-scale B2B bulk buyers, boutique retail partners, and export agents with clearer, standardized product sheets.
- **Standardised Product Metadata: Supports state cooperatives, craft societies, and NGOs with clean, structured, and searchable catalogs of local artisan outputs.**
- Stronger Institutional Market Linkages: Bridges physical-to-digital gaps, allowing marginalized creators to directly service corporate gifting channels and large buyer networks.
- **ONDC-Ready Pathway Layout: Structures inventory output data directly into schema-ready formats to facilitate future frictionless integration on open commerce.**

HOW WE WILL MEASURE PILOT IMPACT (Measurable Success Metrics)

TIME SAVED

Drafting Speed

Catalogue creation cut from days to under 2 mins per item

LISTINGS APPROVED

Artisan Approval

Number of voice-drafted listings reviewed & confirmed

BUYER ENQUIRIES

Market Matching

B2B bulk leads and matches made via regional database

INDEPENDENT USE

Self-Sufficiency

% of rural artisans completing listings with zero tech aid

❖ Research & References

Official sources and statutory structures that informed KarigarSetu's design and digital-market linkage strategy

IndiaHandmade

Ministry of Textiles, Government of India. Official e-marketplace created to support weavers and artisans by enabling direct customer sales.

mindiahandmade.com

BHASHINI

Digital India Initiative • MeiT.Y. Translates and transcribes regional audio, breaking critical language barriers for voice-led listings.

bhashini.gov.in

ONDC (Open Network)

DPIIT, Government of India. Provides a unified, open framework for digital-commerce discovery and frictionless localized merchant access.

ondc.org

DC (Handicrafts)

Ministry of Textiles. Nodal government body directing nation-wide handicraft clusters, artisan welfare, and design training development.

handicrafts.nic.in

Indian Handicrafts Portal

Development Commissioner (Handicrafts). Centralized portal for registering artisan identity, checking schemes, and tracking regional crafts.

indian.handicrafts.gov.in

Amazon Karigar

Amazon India Initiative. Supports Indian artisans and small businesses by providing access to online markets, helping them showcase and sell traditional handmade products to customers nationwide

Amazonkarigar.com

◆ All AI-generated catalogue details remain editable and artisan-approved before publishing — ensuring accuracy, ownership, and trust.